SMARTBRIDGE — Technology Stack

Architecture Components & Tech Choices for MedSafe AI

Date:	August 12, 2026	**Team ID:**	MEDSAFE_AI_01
Project Name:	MedSafe AI — Healthcare Safety & Preventive Awareness Platform	**Max Marks:**	2 Marks

S.No	Architecture Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	**Streamlit (Python)**	Allows rapid prototyping of an interactive, accessible healthcare web UI with clean components for file uploads, score meters, and risk tables.
2	**Backend / Server-Side**	**Python 3.11 + FastAPI**	Provides high-performance asynchronous API endpoints to coordinate OCR processing, fuzzy string matching, and LLM inference seamlessly.
3	**Database / Data Storage**	**SQLite + Local JSON Datasets**	Lightweight, zero-configuration local database to store drug interaction rules and red-flag symptom keywords without cloud dependencies.
4	**Local AI / OCR Engines**	**Ollama (LLaMA 3 8B) + PyTesseract OCR**	PyTesseract extracts text from prescription images, while local LLaMA 3 formats structured JSON and generates plain-language patient advice offline.
5	**Matching & Search Engine**	**RapidFuzz Library**	Enables fast string matching resilience against patient typos or OCR extraction errors (e.g., matching "Aspirne" to "Aspirin").
6	**Deployment & Containerization**	**Docker + Docker Compose**	Ensures single-command deployment across platforms, bundling Streamlit, FastAPI, PyTesseract, and Ollama into a isolated environment.
7	**Version Control & CI/CD**	**GitHub Actions**	Automates code testing, linting, and Docker image build checks across team commits.